Simple models of socio-economic behavior with heterogeneous agents and local interactions

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Adoption of a new technology[1]

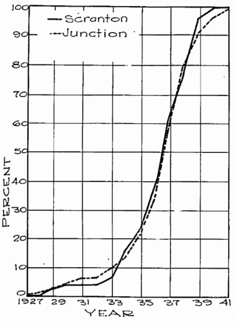
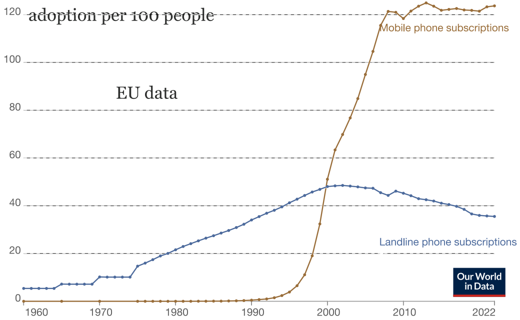


Fig. 1. Cumulative percentages of operators accepting hybrid seed in the two communities during each year of the diffusion process.



The number of adopters follows an **S-shaped curve**.

[1] B. Ryan & N. Gross, *Res. Bull.* (1950); https://ourworldindata.org/

Binary choice models: contagion

Phenomenological explanation (**contagion process**): the new technology is so much better that agents adopt it as soon as they get in contact with some earlier adopter.

The number n_t of adopters grows as (λ contact rate)

$$n_{t+1} = n_t + (N - n_t)\,\lambda n_t/N$$

For larger N, the dynamics is described by the **Bass model** (1969)[2]:

$$\frac{d\rho}{dt} = \lambda\rho\,(1-\rho)$$

where $\rho = n/N$ is the fraction of adopters.

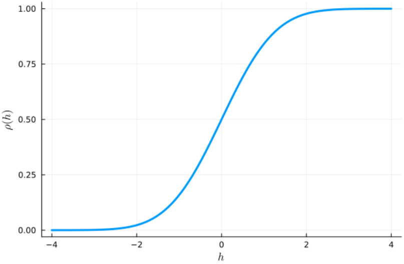
[2] F. Bass, *Man. Sci.* (1969)

Binary choice models: heterogeneity

Adoption driven by **exogeneous factors** (quality/price changes)[3]:

- agents are exposed to a common *incentive field* $h(t) \in \{-\infty, +\infty\}$
- agent have *personal preferences* represented by thresholds $\theta_i \sim g(\theta)$

If $h(t)$ is increased over time, $\rho_{t+1} = \int_{-\infty}^{h(t)} g(\theta)\, d\theta$



The shape of the adoption curve depends on the pdf $g(\theta)$ (here gaussian).

[3] P. Young, *Am. Econ. Rev.* (2009)

Binary choice models: social influence/ imitation

Consider **social influence** with homogeneous agents:

Agent i has utility

$$u_i = ba_i \sum_{j \neq i} a_j - ca_i \sum_{j \neq i} (1 - a_j) \quad = \quad [(b + c)\,\rho - c]\, a_i$$

If every agent adopts ($a_i = 1$) with same probability $\pi_i^t(1) = \rho_t$, then the replicator equation reads

$$\begin{aligned} \frac{d}{dt}\rho_t &= \rho_t \left[\bar{u}_t(1) - \sum_{a' \in A} \pi_i^t(a')\, \bar{u}_t(a') \right] \\ &= \rho_t (1 - \rho_t) \left((b + c)\, \rho_t - c \right) \end{aligned}$$

Two asymptotically stable equilibria at $\rho = 0$ and $\rho = 1$ (with metastability).

Only if $c < 0$, it reduces to the Bass model.

Binary choice models: social influence/ imitation

Ising-Weidlich model: stochastic action revision (SBRD)[4]

$$\Pr\left[a_i^{t+1}=1|a_{-i}^t\right]=\frac{e^{\beta\Delta u_i}}{1+e^{\beta\Delta u_i}}$$

gives action increase with probability

$$\begin{aligned}\Pr\left[N_+\to N_++1\right] &= \frac{1}{N}\sum_i \Pr\left[a_i^{t+1}=1|a_{-i}^t\right]\Pr\left[a_i^t=0\right]\\ &= \frac{e^{\beta U(\rho)}}{1+e^{\beta U(\rho)}}(1-\rho)\end{aligned}$$

Then

$$\begin{aligned}\frac{1}{N}\left(\langle N_+\rangle_{t+1}-\langle N_+\rangle_t\right) &= \frac{e^{\beta U}}{1+e^{\beta U}}(1-\rho)-\frac{1}{1+e^{\beta U}}\rho\\ \Longrightarrow \qquad \frac{d\rho}{dt} &= \frac{e^{\beta(((b+c)\rho-c))}}{1+e^{\beta(((b+c)\rho-c))}}-\rho\end{aligned}$$

[4] W. Weidlich, Brit. J Math Stat. Psyc. (1971); W. Brock & S. Durlauf, Rev. Econ St (2001)

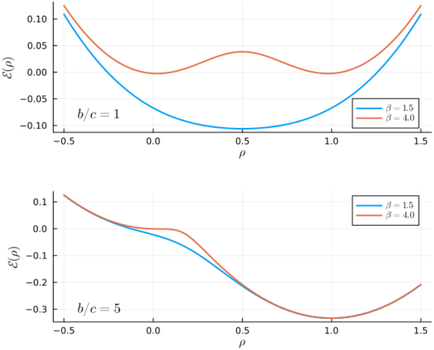
Binary choice models: social influence/ imitation

Potential energy

$$\mathcal{E}(\rho) = \frac{\rho^2}{2} - \frac{\log\left(1 + e^{\beta((b+c)\rho - c)}\right)}{\beta(b+c)}$$

Critical point β_c (for $b = c$, no preference)

$$\mathcal{E}''(\rho) = 0 \implies \beta_c = 1/c$$



Stochasticity make possible to jump from one to the other equilibrium (metastability for $b \gg c$)

For small ρ and $b \gg c$ and $\beta b \gg 1$, it recovers Bass model of adoption (rational agents with strong preference).

Binary choice models: RFIM

Heterogeneity + social influence (Random Field Ising Model)[5]:

$$\begin{aligned} \Delta u_i^t &= u_i^t(1) - u_i^t(0) \\ &= h(t) - \theta_i + \frac{J}{N-1} \sum_{j \neq i} a_j^t \end{aligned}$$

where $J/(N-1)$ replaced $b+c$ and $\theta_i - h(t)$ replaced c.

Upon increasing $h(t)$,

$$\rho_{t+1} = \frac{1}{N} \sum_i \Pr\left[\theta_i < h(t) + J\rho_t\right] = \int_{-\infty}^{h(t)+J\rho_t} g(\theta)\, d\theta$$

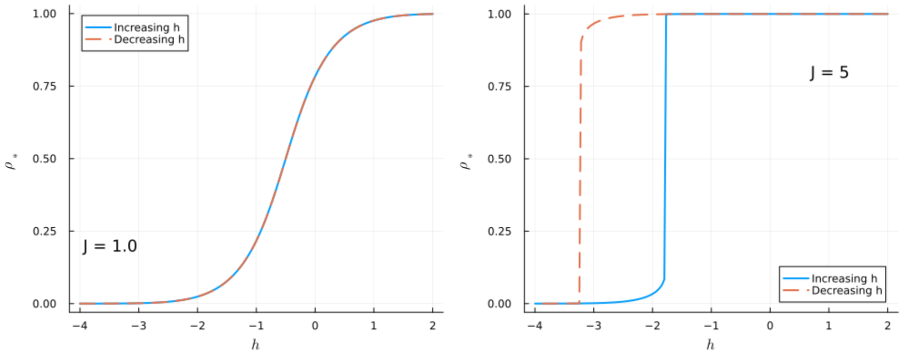
At given h, the fixed point is the solution of

$$\rho_* = \int_{-\infty}^{h+J\rho_*} g(\theta)\, d\theta$$

[5] J.P. Bouchaud, *J Stat. Phys.* (2013); J. Sethna et al. arXiv:cond-mat/0406320 (2004)

Binary choice models: RFIM

Phase transition at J_c: strong imitation leads to coexistence of equilibria and hysteresis



In economic context: the demand for a product can be a multi-valued function of price.[6]

[6] J.P. Nadal et al., *Quant. Fin.* (2006)

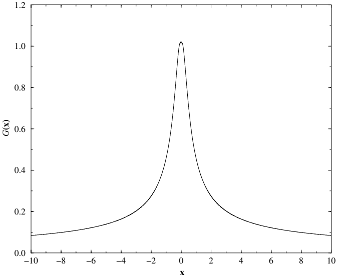
Binary choice models: RFIM

Critical scaling: for $J \to J_c^-$, the "speed of change"[7]

$$\frac{d\phi}{dh} \propto (J_c - J)^{-1} \mathcal{G}\left(\frac{h - h_c(J)}{(J_c - J)^{3/2}}\right)$$

with the scaling function

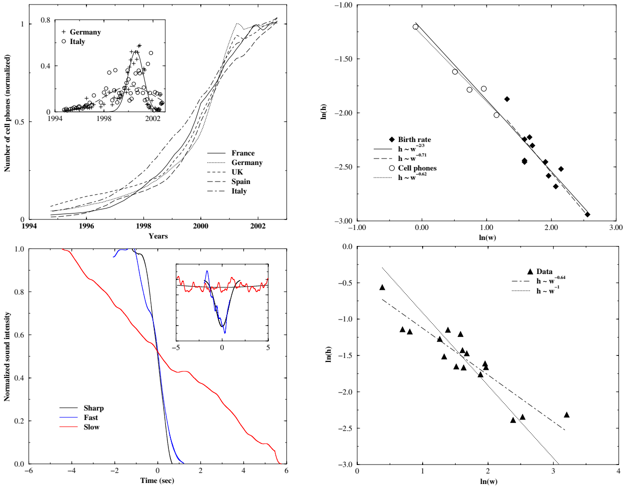
$$\mathcal{G}(x) \sim \begin{cases} \text{const.} & x \to 0 \\ x^{-2/3} & x \to \infty \end{cases}$$



Measuring height and width of the function gives testable predictions!

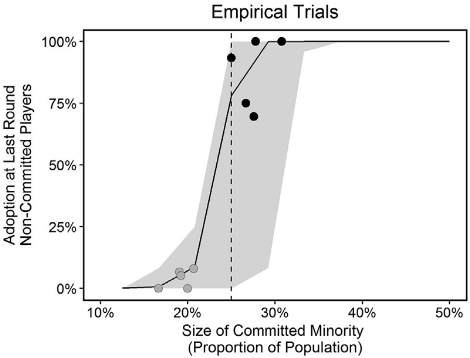
[7] J.P. Bouchaud, *J Stat. Phys.* (2013)

These scaling laws are observed in real-world data[8]



[8] O. Michard & J.P. Bouchaud, *Eur. Phys. J. B* (2005)

RFIM predicts the existence of **tipping points**:[9]



A critical mass effect is observed in many adoption processes, through surveys (left) and experiments (right).

[9] Y. Peng & X. Bai, *Env. Inn. Soc. Trans.* (2024); D. Centola et. al., *Science* (2018)

Binary choice on networks: coordination

Linear Threshold Model[10] on a (undirected) network $G = (V, E)$

- actions $a_i \in \{0, 1\}\,, \forall i \in V$
- utility function

$$u_i\left(a_i, a_{\partial i}\right) = \theta_i k_i \left(1 - a_i\right) + a_i \sum_{j \in \partial i} a_j$$

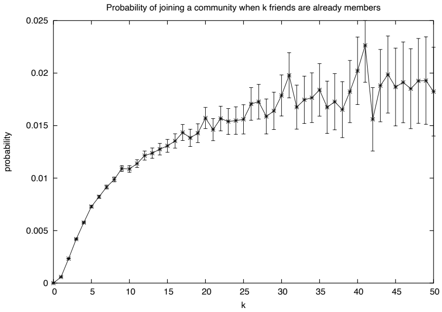
  with local interactions $a_{\partial i} = \left\{a_j \middle| \left(i, j\right) \in E\right\}$
- thresholds (intrinsic resistance) $\theta_i \sim \mathcal{N}\left(\mu, \sigma\right)$
- no external incentive field

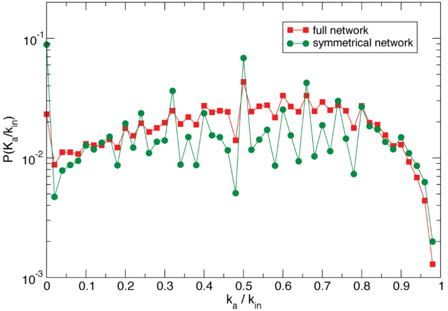
Two versions:

- game of complete information (agents know G and $\vec{\theta}$) $\Longrightarrow$ NE
- game of incomplete information (agent i knows k_i and statistical properties of G and $\vec{\theta}$) $\Longrightarrow$ Bayesian NE

[10] M. Granovetter, *Am. J. Sociol.* (1978); D. Watts, *PNAS* (2002); M. Jackson, *Social and Economic Networks* (2009)

Real-world data:



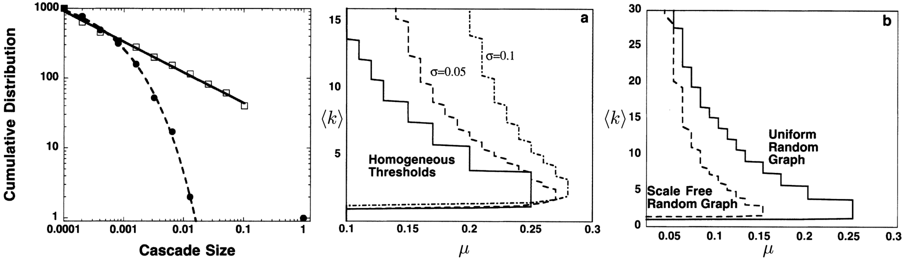


Thresholds as function of number of peers already involved: in an online community (LiveJournal)[11], in Spanish protest mobilization (via Twitter)[12]

[11] L. Backstrom et al. *KDD* (2006)

[12] S. Gonzalez-Bailon et al., *Sci. Rep.* (2011)

Adoption cascades on random graphs only defined by their degree distribution p_k and with gaussian thresholds $\theta \sim \mathcal{N}(\mu, \sigma)$.[13]



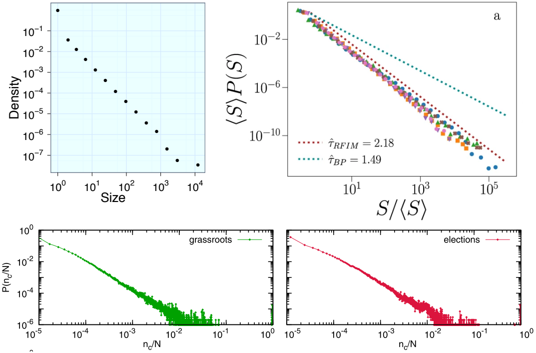
Emergence of cascades with $P(s) \sim s^{-3/2} e^{-s/s_{\mathrm{cutoff}}}$ (alike to RFIM)

$s_{\mathrm{cutoff}} \to \infty$ at the lower critical point, $\langle k \rangle = 1.05$, $\mu = 0.18$

[13] D. Watts, *PNAS* (2002)

Binary choice on networks: coordination

Cascade phenomena are observed in online communities and social media with network structure, such as Twitter (top left/bottom), Weibo, StackOverflow, Parler, Telegram, Delicious (top right)[14].



[14] E. Bakshy et al. *ACM* (2011); D. Notarmuzi et al. *Nat. Comm.* (2022); R. Banos et al. *EPJ Data Sci.* (2013)

Binary choice on networks: coordination

Games of strategic complements satisfy **supermodularity**:[15]

- partial order: $a \geq a'$ iff $a_i \geq a'_i \; \forall i \in V$
- increasing differences: for all $a \geq a'$

$$u_i\left(a_i, a_{\partial i}\right) - u_i\left(a'_i, a_{\partial i}\right) \geq u_i\left(a_i, a'_{\partial i}\right) - u_i\left(a'_i, a'_{\partial i}\right)$$

Every action profile a^* that is fixed-point of the best-response conditions

$$a_i^* = b_i\left(a_{-i}^*; \theta_i\right) = \begin{cases} 1 & \text{if } \sum_{j \in \partial i} a_j^* - \theta_i k_i \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

is a **pure-strategy Nash Equilibrium**.

[15] M. Jackson, *Social and Economic Networks* (2009)

Structure of Nash Equilibria

The set of psNE forms a **complete lattice**: it admits a partial order and every non-empty subset has a supremum and an infimum in the set (Pareto rankable equilibria).

$\Longrightarrow \exists$ minimum NE, maximum NE

There are possibly many (exponentially in N) in between.

- Find *minimum NE*: start from $\vec{0}$ and repeat best response till convergence
- Find *maximum NE*: start from $\vec{1}$ and repeat best response till convergence

With slight modification every psNE can be found.[16]

[16] P. Echenique, *J Econ. Th.* (2007)

Binary choice on networks: coordination

Stability of minimum NE and (adoption) **cascade conditions**:[17]

Forced monotone single-agent deviation, followed by BRD till convergence. If the new NE is $O(N)$ away, the minimum NE is unstable.

On a **random graph** with degree distribution p_k

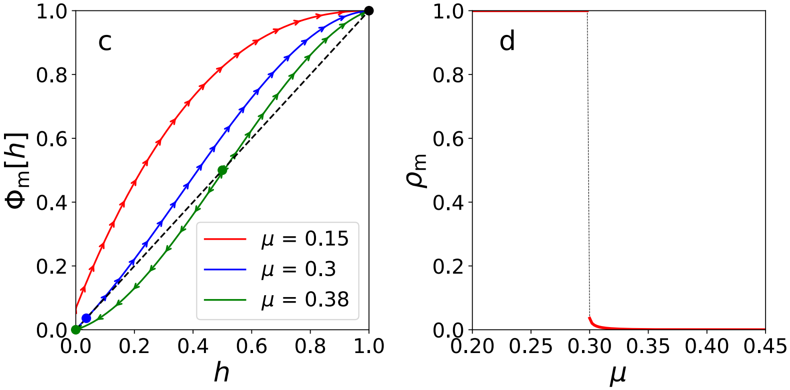
$$
\begin{aligned}
h &= \sum_k \frac{k p_k}{\langle k \rangle} \sum_{m=0}^{k-1} B_{k-1,m}(h) \Pr[\theta \leq m/k] \\
\rho_{\mathrm{m}} &= \sum_k p_k \sum_{m=0}^{k} B_{k,m}(h) \Pr[\theta \leq m/k]
\end{aligned}
$$

the cascade condition ($\exists$ solution with density of adopters $\rho_{\mathrm{m}} \gg 0$) is

$$
\sum_k p_k \frac{k(k-1)}{\langle k \rangle} \Pr[\theta \leq 1/k] > 1
$$

[17] D. Watts, *PNAS* (2002); J.P. Gleeson et al., *Phys Rev E* (2007); L. Dall'Asta, *JSTAT* (2021)

For instance, for a random regular graph of degree $K = 4$, and gaussian thresholds $(\sigma = 0.1)$[18]



Are there other equilibria? Can we find them?

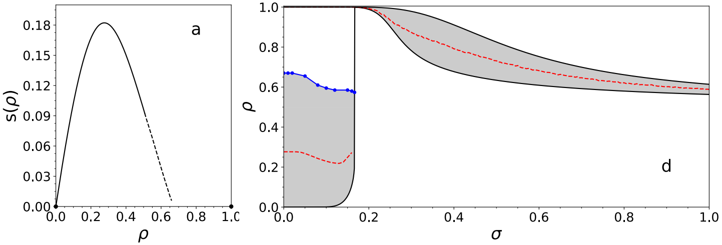
[18] L. Dall'Asta, *JSTAT* (2021)

Statistical mechanics methods can be used to characterize NE[19]

Even for a random regular graph, for large N, the number of NE can be exponential in N

$$\mathcal{N}_{NE}(\rho) \sim e^{Ns(\rho)}$$

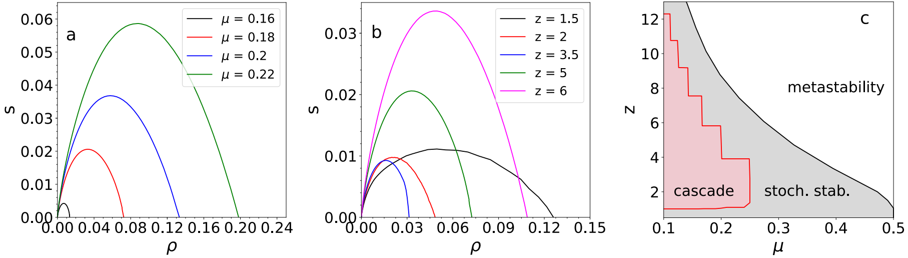
where $s(\rho) \geq 0$ **entropy** function. Here for $K = 4$, $\mu = 0.35$, when $\vec{0}$ is stable, a large spectrum of equilibria exists



[19] L. Dall'Asta, *JSTAT* (2021)

Binary choice on networks: coordination

In general networks, the equilibrium spectrum exists independently of the cascade boundaries (here ER random graphs)



In the cascade region, these NE are unstable to BRD perturbations (in favor of the efficient equilibrium).

The efficient equilibrium is also stochastically stable in a wider region.

Binary choice on networks: coordination

Stochastic stability: agents update their actions with probability (SBRD)

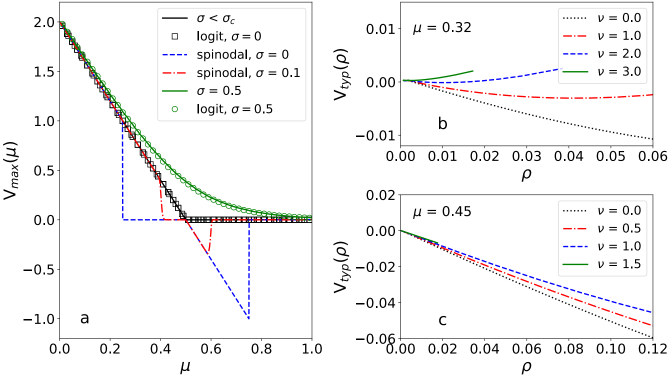
$$\Pr\left[a_i^{t+1} = a_i | a_{\partial i}^t\right] = \frac{e^{\beta u_i\left(a_i, a_{\partial i}^t\right)}}{\sum_{a_i'} e^{\beta u_i\left(a_i', a_{\partial i}^t\right)}}$$

Nash Equilibria: local maxima of the potential function

$$V\left(a;\theta\right) = \sum_{(i,j)\in E} a_i a_j - \sum_i \theta_i k_i a_i$$

Stochastically stable states: global maxima of $V\left(a;\theta\right)$

Binary choice on networks: coordination

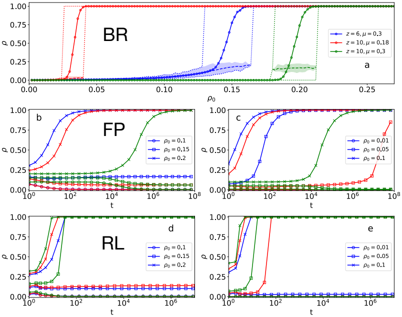


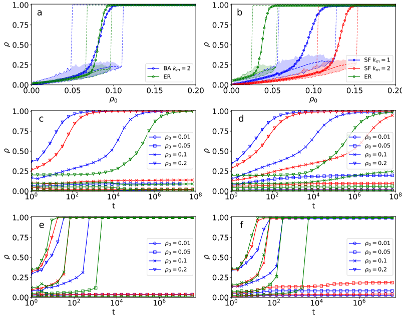
The efficient NE is stochastically stable up to a threshold. The minimum NE can be locally stable (*metastability*).

It exists a region (for $\sigma > 0$) in which the minimum NE is not even locally stable but cascade phenomena are prevented by entropic reasons.

Binary choice on networks: coordination

The structure and stability of NE influences equilibrium selection and learning dynamics (Left: ER graphs; Right: power-law graphs)





Dynamics with strong memory or low stochasticity can be trapped in low efficiency equilibria.

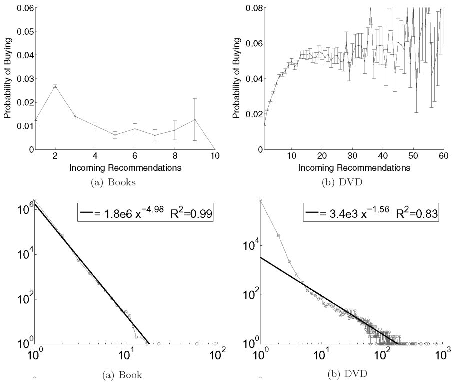
Binary choice on networks: coordination

Experimental (and numerical) evidences:[20]

- higher density, small-world property and clustering favor convergence to efficient equilibria
- local interactions favor convergence to stochastically stable equilibria
- degree heterogeneity promote coordination, hubs trigger adoption cascades
- coordination is more difficult when the network exhibits community structure and cliquishness
- inefficient equilibria preferentially involve low-degree nodes

[20] S. Berninghaus, et al., *Game Econ Behav* (2002); A. Cassar, *Game Econ Behav.* (2007); C. Keser, *Econ Lett* (1998); K.B. My, et al., *J Evol Econ* (1999); S. Judd, et al. *PNAS* (2010).

Viral marketing:

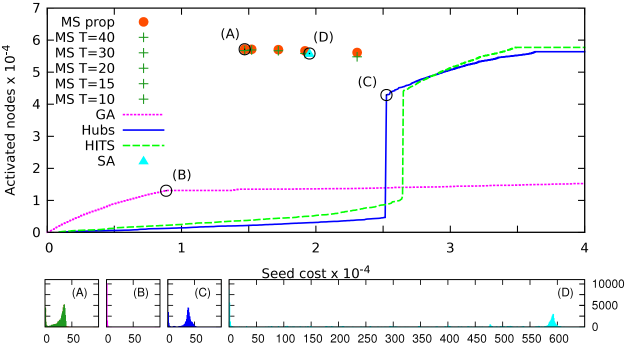


Data[21] suggest a more complex social-influence model (too many recommendations are deleterious).

[21] J. Leskovec et al., *ACM Trans.* (2007)

Binary choice on networks: coordination

However, given the microscopic model, **optimal seeding** can be obtained using stat-mech algorithms[22]



(Top) Optimal spread on Epinions network with thresholds proportional to the in-degree. (Bottom) Number of newly activated nodes over time.

[22] F. Altarelli et al., JSTAT (2013)

Binary choice on networks: local public goods

Public Goods game on a (undirected) network $G = (V, E)$

- actions $a_i \in \{0, 1\}$, $\forall i \in V$
- utility function

$$u_i(a_i, a_{\partial i}) = \begin{cases} 1 - c & \text{if } a_i = 1 \\ 1 & \text{if } a_i = 0, \qquad \prod_{j \in \partial i} (1 - a_j) = 0 \\ 0 & \text{if } a_i = 0, \quad a_j = 0, \forall j \in \partial i \end{cases}$$

with $c \in (0, 1)$.

Two versions:

- game of complete information (agents know G) $\Longrightarrow$ NE
- game of incomplete information (agent i knows k_i and statistical properties of G) $\Longrightarrow$ Bayesian NE

Binary choice on networks: local public goods

Games of strategic substitutes satisfy **submodularity**:[23]

- partial order: $a \geq a'$ iff $a_i \geq a'_i \; \forall i \in V$
- decreasing differences: for all $a \geq a'$

$$u_i\left(a_i, a_{\partial i}\right) - u_i\left(a'_i, a_{\partial i}\right) \leq u_i\left(a_i, a'_{\partial i}\right) - u_i\left(a'_i, a'_{\partial i}\right)$$

Every action profile a^* that is fixed-point of the best-response conditions

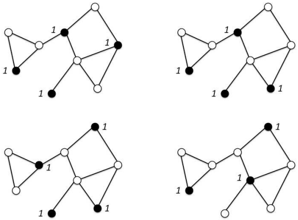
$$a_i^* = b_i\left(a^*_{-i}; \theta_i\right) = \prod_{j \in \partial i} \left(1 - a_j^*\right)$$

is a **pure-strategy Nash Equilibrium**.

[23] M. Jackson, *Social and Economic Networks* (2009)

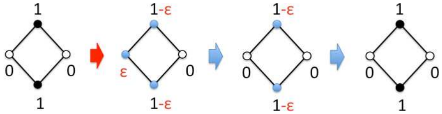
Binary choice on networks: local public goods

Pure-strategy NE are the maximal independent sets of the graph[24]



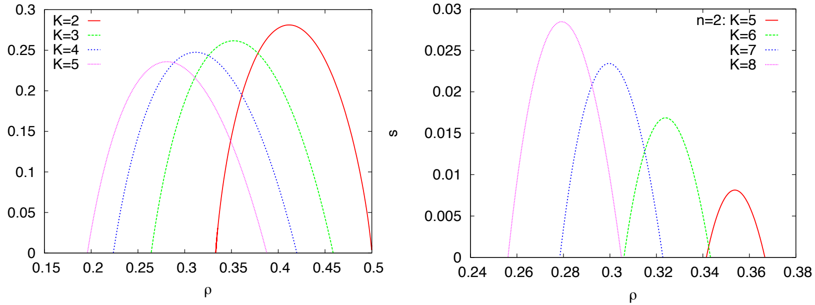
NE differ in the number of agents contributing (free-riding problem).

Stable NE (at least 2 contributors around every free-rider):



[24] Y. Bramoullé et al., *J Econ Th* (2007)

Entropy of psNE[25]



Finding a NE is simple (right), but finding stable ones (right) is not (stronger correlations).

Similarly, finding NE with a given density ρ of contributors is hard.

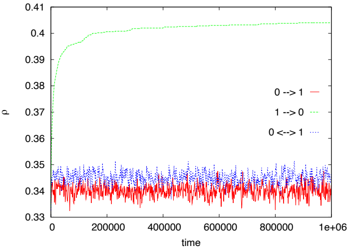
[25] L. Dall'Asta et al. *Phys Rev E* (2010)

It exists a potential function

$$V = \sum_i \left(a_i - \frac{1}{2} a_i^2 \right) - \frac{1}{2} \sum_{(i,j)} a_i a_j = \frac{1}{2} n_1 - n_{11}$$

Stochastically stable states are NE with maximum density.[26]

An incentive mechanism: perturbed BRD: $0 \to 1$ (red), $1 \to 0$ (green), and both (blue). Fraction of stable free-riders (right).



[26] L. Boncinelli et al., *Game Econ Behav* (2012)

General model with continuous actions $a_i \in \mathbb{R}$ [27]

$$u_i\left(a_i, a_{-i} \middle| G\right) = a_i \left(b_i + \beta \sum_{j \in V} g_{ij} a_j\right) - \frac{1}{2} a_i^2 + P_i\left(a_{-i}, g, b\right)$$

where $g = \left(g_{ij}\right)_{i,j \in \mathcal{N}}$ is a weighted adjacency matrix and $\beta > 0$ ($\beta < 0$) for complements (substitutes).

Any NE satisfies $\left[I - \beta g\right] a^* = b$

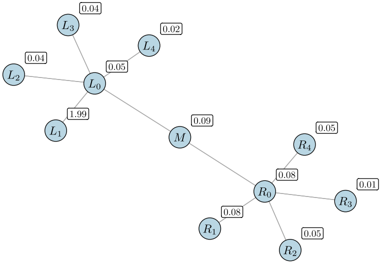
Planner's problem: distribute incentives $\delta b = b' - b$ such that

- maximize welfare $\max_{b'} \sum_{i \in \mathcal{N}} u_i$
- subject to (unique) NE constraint $a^* = \left[I - \beta g\right]^{-1} b'$
- budget constraint $\sum \delta b^2 \leq B$.

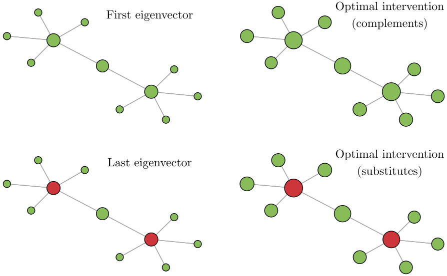
[27] A. Galeotti et al., *Econometrica* (2020)

Binary choice on networks: targeting interventions

Optimal intervention with large budgets



(A) The network and status quo standalone marginal returns

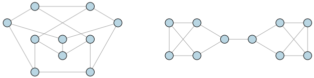


(B) Eigenvectors and interventions

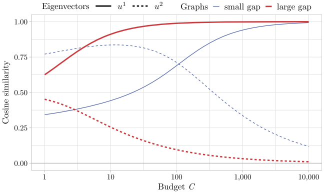
$\delta b > 0$ (green) and $\delta b < 0$ (red) resemble first and last eigenvectors of the adjacency matrix.

Binary choice on networks: targeting interventions

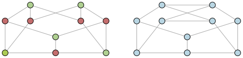
Optimal intervention at different budgets



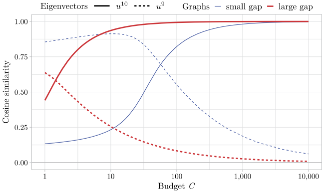
(A) Large (left) and small (right) spectral gap



(B) Cosine similarity and strategic complements



(C) Large (left) and small (right) bottom gap



(D) Cosine similarity and strategic substitutes

Optimal intervention converges to the topological one for large budgets.

Binary choice on networks: collaboration

Consider a game of local contribution[28]

$$u_i\left(a_i, a_{-i}\right) = -X_i a_i + \sum_{j \in \partial i} a_j$$

where defection $(a_i = 0)$ is a dominant strategy for all $X_i > 0$.

The stage game admits a unique NE in which all agents defect.

The repeated game with utility

$$U_i\left(\vec{s}_i, \vec{s}_{-i}\right) = (1-\delta) \sum_{t \geq 0} u_i\left(a_i^t, a_{-i}^t\right) \delta^t$$

is studied when only trigger strategies are admitted.

Does punishment credibility depend on local network structure?

[28] L. Dall'Asta et al., *PNAS* (2012)

Binary choice on networks: collaboration

3 players: if one deviates, what should the other two do?

- If the deviate, they get 0,
- If continue collaborating, they with get $1 - X_i$.

So the threat of punishment is not credible if $X_i < 1$.

A credible punishment must be (1) player-specific, (2) reciprocal.

Definition

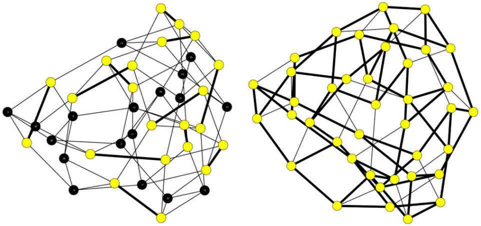
In a **conditional trigger strategy** $\tau_i(\Delta_i)$, player i collaborates ($a_i = 1$) conditional on the collaboration of a subset $\Delta_i \subseteq \partial i$. Let Γ_i be the set of *direct punishers* of i, i.e. $\Gamma_i = \{j \in \partial i : i \in \Delta_j\}$, and call C the subset of agents that play the conditional trigger strategy, while players $i \notin C$ are agents that always defect.

Binary choice on networks: collaboration

Proposition (Collaborative Equilibrium)

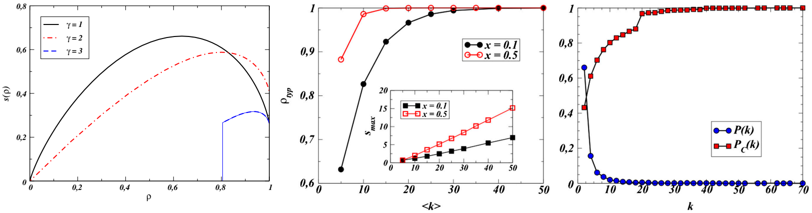
Consider the conditional trigger strategy profile $\tau = \{\tau_i(\Delta_i)\}_{i \in C}$ such that $\forall i \in C$, $X_i \leq |\Gamma_i| < X_i + 1$ and $\Gamma_i = \Delta_i$. If for all $i \in C$ and for all $j \in \Gamma_i$ it also holds that $D_{j \to i} = D_{i \to j}$, then the set C of players endowed with the strategy profile τ defines a Nash equilibrium of the repeated game for sufficiently large $\delta < 1$.

Example of collaborative equilibria (collaborators in yellow, defectors in black) with $0 < X_i < 1$, $\forall i$ (left) and $2 < X_i < 3$, $\forall i$ (right).



Binary choice on networks: collaboration

Statistical mechanics methods can be used to determine properties of Collaborative Equilibria.

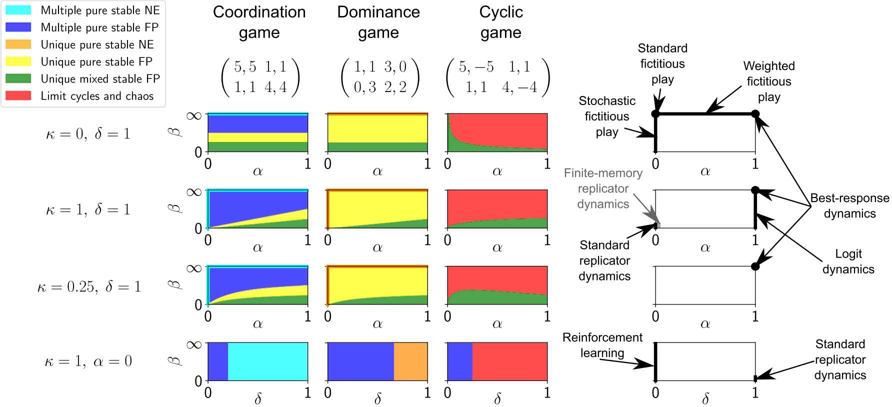


- CEs do not exist for every density ρ of collaborators (critical mass)
- Denser networks favor collaboration
- Hubs are more likely to collaborate (unless star-like nets)

Theory corroborates experimental results.

Complex games and learnability

Taxonomy of 2×2 games: convergence of Reinforcement Learning (EWA algorithm) depending on the parameters (memory loss α, experience discount κ, stochasticity β) and payoff matrices:[29]



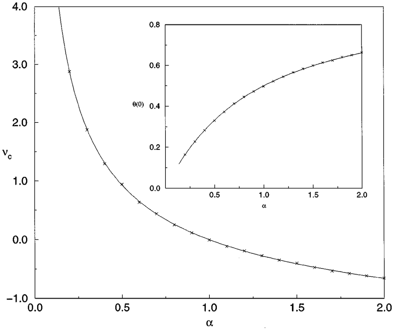
29 M. Pangallo et al., *Game Econ Behav* (2022)

Complex games and learnability

Large two-player games with random (gaussian) payoff matrices:[30]

Player 1 has N actions, player 2 has M actions.

In the limit $N, M \to \infty$ with $M/N = \alpha$

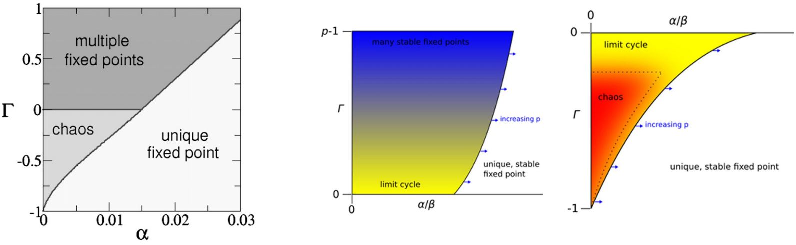


Value of the game (main), fraction of actions played (inset).

[30] J. Berg & A. Engel, *Phys Rev Lett* (1998)

Complex games and learnability

Two-player and multi-player games with random payoff matrices (gaussian with non-zero covariance Γ) and reinforcement learning dynamics[31]



Left: Memory loss α and correlations Γ (at fixed β) prevent to converge to steady-state mixed strategy. Centre-Right: Same for many p players, the u.f.p. region shrinks

Complex non-equilibrium behavior is the norm in large multi-agent systems.

[31] T. Galla & D. Farmer, *PNAS*, 2013; J. Sanders et al., *Sci. Rep.* (2018)

Complex games and learnability

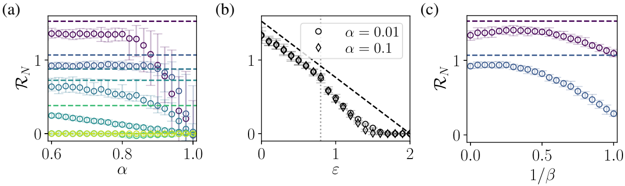
N-player game inspired by the physics of disordered systems:[32]

Utility function $u_i^t = \frac{a_i^t}{2}\left(h_i^t + \sum_j J_{ij} a_j^t\right)$ with $a_i^t \in \{\pm 1\}$,

$J_{ij} = \left(1 - \frac{\epsilon}{2}\right) J_{ij}^S + \frac{\epsilon}{2} J_{ij}^A$, $J_{ij}^S, J_{ij}^A \sim \mathcal{N}(0, \sigma)$

Q-learning dynamics defined by $\Pr\left[a_i^t = \pm 1\right] \propto e^{\pm \beta Q_i^t}$ and

$$Q_i^{t+1} = (1-\alpha)\, Q_i^t + \alpha \left[u_i^t\left(a_i = +1, a_{-i}^t\right) - u_i^t\left(a_i = -1, a_{-i}^t\right)\right]$$



Learning either gets trapped by suboptimal fixed points or enters never-ending chaotic or quasiperiodic evolution.

[32] J. Garnier-Brun et al., *Phys Rev X* (2024)